**Course Code: ISC – S101** **Breakup: 3 – 0 – 3 – 5**
**Course Name: Programming & Computing(C & UNIX)**

**Course outcomes (CO):** At the end of the course, the student will be able to:

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|---|---|
| CO1 | Recollect various programming constructs and to develop C programs |
| CO2 | Understand the fundamentals of C programming |
| CO3 | Choose the right data representation formats based on the requirements of the problem |
| CO4 | Implement different Operations on arrays, functions, pointers, structures, unions and files |

**Course Details:**
Basic concepts of Computers, Basic UNIX Concepts and Vi - Editor
Introduction to C: Basic Programming concepts, Program structure in C, Variables and Constants, Data types, Conditional statements, control statements, Functions, Arrays, Structures, Introduction to pointers and Introduction to File Systems.

**Text Books and References:**


1. Programming in C, Schaum Series, 3rd edition, BPB Publication, Byron S. Gottfried
2. The 'C' Programming, Denis Ritchi, Second edition, PHI, 1988
3. Mastering C, Venugopal, Second edition, TMH, 2006
4. Let Us C, Yashavant Kanetkar, 18th Edition, BPB, 2021
5. Programming in ANSI C, Balaguruswami, Eighth Edition, TMH, 2019


**Computer Programming Lab:**

Learning OS Commands
Practice of all Internal and External DOS Commands, Writing simple batch programs, Exposure to Windows environment, Practice of UNIX commands and Vi editor, Writing simple shell script
C Programming:
Practicing programs to get exposure to basic data types, algebraic expressions, Conditional statements, Input Output Formatting, Control structures, arrays, functions, structures, pointers and basic file handling